UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lab 5 – Audiometer

EIN3241: Human Factors and Ergonomics
Spring 2026

March 6, 2026

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Admin Issues

- Lab 4 due TODAY
- Midterm:
  - Monday October 20
  - 8:20 PM – 10:10 PM in FLG 270
  - Up until Lecture 5E (including Labs 1,2,3, and 4).
- Today's Lab is not due until March 23.

outline

- Lab Objectives
- AMBCO Audiometer
- Experiment
  - Equipment
  - Data Collection Procedure
- Operation Instructions

Introduction

- Lab Objectives
  - To be able to conduct hearing measurement
  - To be familiar with audiometer control and their functions

AMBCO Audiometer

- Audiometer:
  - Measures hearing threshold (or hearing capability) of a person at different frequency levels.
  - Hearing threshold refers to the lowest intensity level at which a person responds approximately 50% of the time
  - Hearing threshold is expressed in the decibel unit (dB)

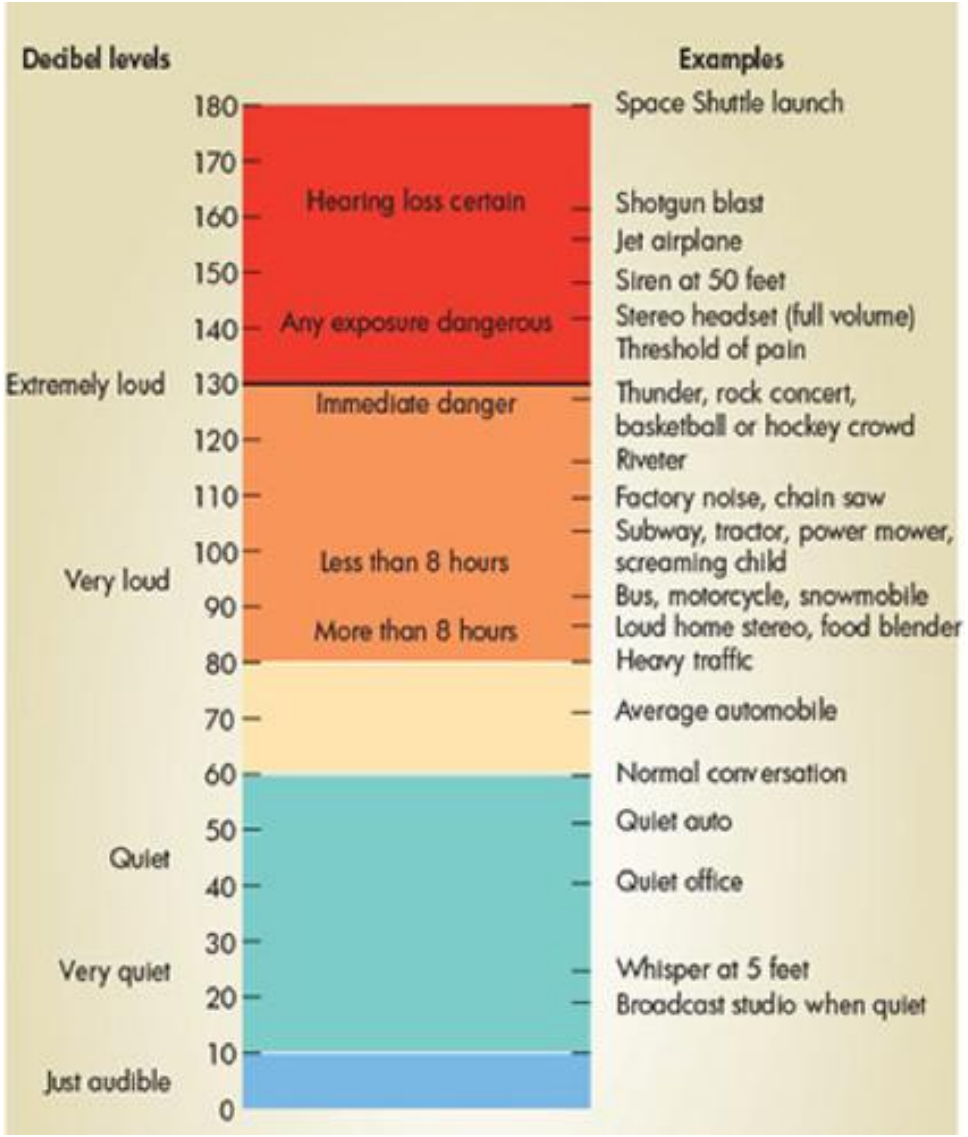
Experiment

- **Equipment**:
  - 3 AMBCO Audiometers

Experiment

- 10 people will be asked to be volunteers for this experiment
- Th experimenter will select different frequencies and measure the hearing threshold
  - 250 Hz
  - 2000 Hz
  - 4000 Hz
  - 6000 Hz
  - 8000 Hz

Sound Scale

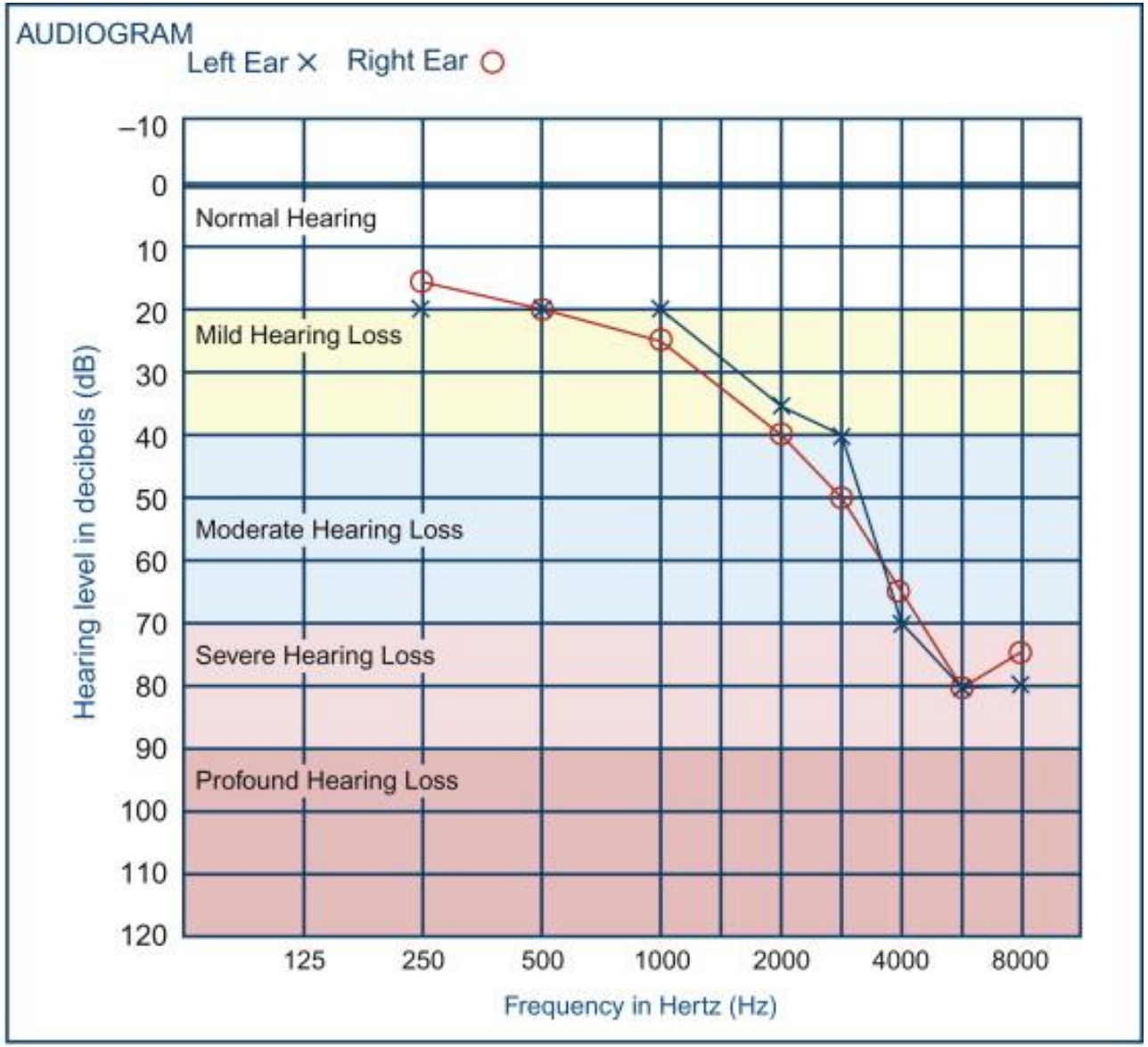

Decibel levels
Examples
180
170
160
150
140
130
120
110
100
90
80
70
60
50
40
30
20
10
0
Extremely loud
Very loud
Quiet
Very quiet
Just audible
Hearing loss certain
Any exposure dangerous
Immediate danger
Less than 8 hours
More than 8 hours
Space Shuttle launch
Shotgun blast
Jet airplane
Siren at 50 feet
Stereo headset (full volume)
Threshold of pain
Thunder, rock concert, basketball or hockey crowd
Riveter
Factory noise, chain saw
Subway, tractor, power mower, screaming child
Bus, motorcycle, snowmobile
Loud home stereo, food blender
Heavy traffic
Average automobile
Normal conversation
Quiet auto
Quiet office
Whisper at 5 feet
Broadcast studio when quiet

Data Collection Procedures

1. Participant (P) puts earphones on with the **red** side over the **right** ear and the **blue** side over the **left** ear.
2. P should pay close attention to the tones as some will be very faint and difficult to hear.
3. P must not see the manipulation of the controls during testing.
4. First, test the **right ear**: the experimenter (E) will select the right ear switch on the audiometer and set the frequency (e.g., 1000 Hz) and the intensity to 50 dB.
5. Then press the tone button on the audiometer for 1 second. E should not use rhythmical presentations of the tones and should avoid giving any clues that the tone has been presented.
6. When P hears the tone, they should press the red response button.

Data Collection Procedures

7. If P hears the tone, lower the intensity level in 10 dB decrements, until P does not hear the tone for three consecutive times out of four.
8. Then, INCREASE the intensity level by increments of 5 dB. At each level, present the tone four times.
9. If the participant hears the tone at least two times, then this intensity level is the hearing threshold (at the frequency tested).
10. E shall record the threshold at this frequency on the audiogram table (like in the Canvas excel sheet)
11. Repeat the above procedures at other frequencies
12. Repeat the above procedures for the **left** ear.



* An example of presbycusis (sloping high-frequency hearing loss).

Hearing Threshold Analysis

- General guideline to compare hearing test results

Average Threshold Level (dB)	
0-15 dB	Normal Hearing
16-25 dB	Slight Hearing Loss
26-40 dB	Mild Hearing Loss
41-55 dB	Moderate Hearing Loss
56-70 dB	Moderately Sever Hearing Loss
71-90 dB	Severe Hearing Loss
>91 dB	Profound Hearing Loss

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LEADING THE CHARGE, CHARGING AHEAD